

M2 – Scientific Contextualization and Tradeoff Analysis:

Low Power Scheduling for High-Level Synthesis

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MILESTONE 2: CONTEXTUALIZATION AND TRADEOFF ANALYSIS

1. Related approaches and where this work sits

Three main strategies exist for reducing power during HLS. Rettberg et al. [1] use *CDFG partitioning*: they group mutually exclusive execution paths into switchable partitions controlled by guard signals. Musoll and Cortadella [2] take a different route: they reduce *operand switching activity* at functional unit inputs during scheduling and resource binding, without restructuring the graph. Shiue and Chakrabarti [3] use *Integer Linear Programming (ILP)* to simultaneously minimise resources, peak power, and peak area during scheduling, plus a separate LP model for resource binding. Khouri et al. [4] address *control-flow intensive (CFI)* circuits specifically, combining scheduling, module selection, resource sharing, and multiplexer network restructuring in one iterative system called IMPACT.

2. Competing and neighboring solution strategies

- **Partitioning with guards** (Rettberg et al.) – targets the structure of the CDFG; works at the graph level before individual operations are bound to resources.
- **Operand activity minimization** (Musoll & Cortadella) – works after scheduling, at the resource binding level; does not change graph structure.
- **ILP-based scheduling** (Shiue & Chakrabarti) – finds an optimal or near-optimal schedule with respect to multiple objectives; computationally heavier than heuristics.
- **Iterative improvement on CFI circuits** (Khouri et al.) – handles control-flow intensive designs holistically; addresses multiplexer networks which account for over 40% of power in CFI circuits.
- **Clock gating / power gating** – technique applied at RTL or gate level rather than HLS; complementary to all approaches above but operates at a lower abstraction level.

3. Strength of related work

Related work on low-power HLS is moderately strong but fragmented. Each paper targets a different part of the HLS flow (scheduling, binding, graph partitioning, or multiplexer

networks) and uses different benchmarks, making direct comparison difficult. There is no unified framework that combines all techniques. More recent work on FPGA-based HLS and machine-learning-driven power estimation exists but is only indirectly related to the scheduling focus of this topic.

4. Tradeoff analysis

TABLE I
 TRADEOFFS ACROSS LOW-POWER HLS APPROACHES

Approach	Strength	Weakness	Power saving
Rettberg et al. [1]	Graph-level; works on control-dominated designs	NP-hard search; overhead; clique guard one benchmark	15%
Musoll & Cortadella [2]	Simple heuristic; low runtime overhead	Only targets functional units; no graph restructuring	Up to 45%
Shiue & Chakrabarti [3]	Jointly optimises power, area, and resources; flexible weights	ILP is NP-hard; slow for large designs	Up to 34%
Khouri et al. [4]	Holistic CFI system; handles multiplexer networks	High complexity; requires full behavioural simulation	Up to 85%

Key tradeoffs across these approaches:

- **Optimality vs runtime:** ILP (Shiue) gives the best formal guarantees but is slow. Heuristics (Musoll, Rettberg) are faster but may miss optimal solutions.
- **Power vs area:** Musoll & Cortadella note that reducing power in resource binding sometimes requires more registers and multiplexers, increasing area.
- **Power vs latency:** Musoll & Cortadella trade latency for operand reutilisation; adding cycles saves power but may violate timing constraints.
- **Scope of reduction:** Rettberg targets dynamic power only. Static leakage is only partially addressed through resource sharing across all approaches.

- **Control vs data dominated:** Rettberg and Khouri target control-flow intensive designs. Musoll and Shiue focus more on data-path intensive benchmarks.

5. *Embedded systems consequences*

- **Energy:** All approaches reduce dynamic power; this directly extends battery life in IoT and wearable devices without changing the algorithm.
- **Timing:** Latency penalties from operand reutilisation or early control scheduling can violate real-time deadlines in hard real-time systems.
- **Memory:** ILP models and compatibility graphs grow with design size; memory may be a constraint on resource-limited synthesis tools for embedded targets.
- **Determinism:** Partitioning and guard logic add control paths; these must be verified to not introduce non-deterministic behaviour in safety-critical systems.
- **Deployment:** All approaches produce standard RTL output, so they are compatible with existing ASIC and FPGA design flows without tool changes.

6. *Why some approaches are included or excluded*

The three companion papers were chosen because they target the same problem level (HLS scheduling and binding) and use overlapping benchmarks (DFGs, CDFGs, DSP kernels). Clock gating at RTL level and DVFS are excluded because they operate at a lower abstraction level and are complementary rather than competing approaches.

REFERENCES

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